

Measuring the Benchmark, Not the Model: Instruction Information Content Confounds Word-Class Grounding Claims in Vision-Language-Action Models

Anonymous ACL submission

Abstract

Vision-Language-Action (VLA) models map a natural-language instruction and a visual observation to robot actions, but it remains unclear how much they rely on the language rather than on visual priors and memorized task layouts. We study this with a *scene-fixed* causal evaluation that holds the initial simulator state constant and perturbs only the instruction, on LIBERO-Goal, across three architectures spanning a $16\times$ parameter range. Language is causally necessary and scale-invariant: blanking the instruction or replacing it with length-matched nonsense collapses success at every scale.

Our main contribution is negative. The widely reported reading of such results—that VLAs bind object nouns strictly but treat action verbs as near-optional—does not follow, because it is confounded by the information structure of the benchmark. In LIBERO-Goal the noun set uniquely identifies all ten tasks ($H(\text{task} \mid \text{nouns}) = 0$ bits) while the verb leaves 1.55 of 3.32 bits unresolved, so conditional on the nouns the verb is *fully redundant* and ignoring it is optimal rather than a grounding failure. Seven of the ten standard verb substitutions preserve the noun set, so the probe cannot detect verb-insensitivity at all; and the apparent “collapse” under object substitution is compliance being mis-scored. A word-class *removal* test confirms the account: deleting the verb is indistinguishable from no edit (McNemar $p = 0.34$) while masking the nouns flips 148 of 200 matched pairs. Isolating the three tasks with true antonym pairs reverses the usual reading—OpenVLA-7B loses 31.7 points ($p < 10^{-5}$) where the uninformative substitutions move it not at all ($p = 0.52$)—so verb-insensitivity survives for exactly one of our three models. We argue that instruction-level information content must be reported before grounding claims are drawn from perturbation studies.

1 Introduction

Vision-Language-Action (VLA) models extend large vision-language backbones with an action head, promising robots that follow free-form natural-language instructions (Brohan et al., 2022, 2023; Driess et al., 2023; Kim et al., 2024; Shukor et al., 2025). Their headline capability is *language conditioning*: the same policy should perform different manipulations depending on what it is told. Yet standard benchmarks report a single aggregate success rate, which cannot distinguish a policy that reads the instruction from one that has memorized a visually cued task layout. Because instructions and scenes are correlated in most benchmarks—the bowl-on-the-stove task looks different from the open-the-drawer task—a model can score well while largely ignoring language.

We isolate the language channel with a *scene-fixed* causal probe. The LIBERO-Goal suite (Liu et al., 2023) shares one object inventory and layout across its ten tasks, which differ only in the goal to be achieved, so a policy cannot infer the task from the scene as it can where each task has its own objects. Holding the initial simulator state strictly constant ($S_0 = \text{const}$) and perturbing *only* the instruction string, we measure how behavior changes under structural corruptions (blanking, length-matched nonsense), semantic substitutions (swapping the object noun, the action verb, or the whole task), and natural paraphrases. This turns a correlational question (“does success correlate with language?”) into a causal one (“does changing the language change the behavior, with everything else fixed?”).

Evaluating three architectures spanning $16\times$ in scale, we find that language is causally necessary and scale-invariant: destroying the instruction destroys success at every scale. This cuts against a rapidly forming consensus that VLAs are

“instruction-blind” (Fei et al., 2025; Zhou et al., 2025; Hou and Zhao, 2026; Zhang and Bisk, 2026; Grant et al., 2026), and we think both are right about their own settings—Goal is the LIBERO suite that places heterogeneous goals in a *shared* environment, so it is the one where a policy cannot succeed by recognising the layout. Which regime a benchmark is in is a property one can compute in advance, and mostly nobody does.

At first sight our results also reproduce the asymmetry that such probes usually report: object swaps collapse success while verb swaps are tolerated, inviting the conclusion that VLAs bind nouns strictly and treat verbs as near-optional. We set out to explain that gap mechanistically and instead found that it is largely an artifact of the measurement. Verbs in LIBERO-Goal carry no task information once the nouns are known, so ignoring them is optimal rather than deficient; most verb substitutions leave the task identity intact and so cannot test verb grounding; and most of the object-swap “collapse” is the policy correctly executing the instruction it was given, scored against the task it was not. The residual effect is real but much narrower than the literature’s framing suggests.

The methodological point generalises beyond this suite: instruction perturbation silently assumes that the perturbed word classes are informationally comparable, and where they are not, the probe measures the benchmark rather than the model.

Contributions

- **Instruction information content as a confound.** Conclusions about which word classes a VLA grounds are not identifiable from perturbation studies without first measuring what each class contributes to task identity. On LIBERO-Goal nouns carry all 3.32 bits and verbs none conditional on them, so verb-insensitivity is optimal and the standard verb-swap probe is uninformative by construction (Section 4.1).
- **A removal test that separates the two.** Because substitution mostly leaves the task unchanged, we remove classes outright: deleting the verb (0 bits) is free, masking the nouns (3.32 bits) is catastrophic—two edits of similar surface size, two orders of magnitude apart in effect (Section 4.2).
- **Re-analysis of the “asymmetric grounding**

gap”. Both halves dissolve. Object-swap “collapse” is largely *compliance*: where the swap yields another real task’s instruction, all three models perform that task and never the original (17/20, 18/20, 20/20). Verb-swap “tolerance” is mostly measured on substitutions that leave task identity intact; splitting them reverses the conclusion for two of three models (Section 4.4).

- **A scene-fixed protocol with an automated invariant check** that fails loudly, motivated by finding the invariant silently broken by ordinary evaluation practice (Section 4.3), plus a kinematic read-out of *when* behaviour diverges (Section 4.6).

2 Related Work

Vision-Language-Action models. Instruction-conditioned manipulation policies scale vision-language pretraining to robot control. RT-1 and RT-2 (Brohan et al., 2022, 2023) and PaLM-E (Driess et al., 2023) established transformer policies that ingest language and pixels; OpenVLA (Kim et al., 2024) open-sourced a 7B Prismatic-backbone policy with discretized action tokens, and SmolVLA (Shukor et al., 2025) showed a compact (< 1B) alternative. We study these families directly, adding OpenVLA-OFT (an L1-regression continuous-action fine-tune), to probe whether language grounding changes with scale or action-head design.

Benchmarks and robustness. LIBERO (Liu et al., 2023) provides suites of language-conditioned manipulation tasks; we use LIBERO-Goal precisely because its shared initial scene lets us isolate the language channel. LIBERO-Para (Kim et al., 2026) measures paraphrase robustness and introduces the PRIDE metric we adopt in Section 4.5; LIBERO-Plus (Fei et al., 2025) perturbs seven visual and physical factors; and Pang et al. (2025) show that OpenVLA succeeds mainly at object positions seen in training, which is a warning that apparent competence on these suites can rest on memorisation.

The instruction-blindness consensus. A striking convergence has formed around the claim that VLAs largely *ignore* language. LIBERO-Plus finds models insensitive to instruction variation while highly sensitive to viewpoint and ini-

tial state. LIBERO-PRO (Zhou et al., 2025) reports outputs that “remain unchanged even when given corrupted instructions or even messy tokens” and attributes this to rote memorisation of action sequences and layouts; LangGap (Hou and Zhao, 2026) reports the same for models scoring above 95% on standard benchmarks and argues that LIBERO’s one-task-per-layout design never forces language use. Zhang and Bisk (2026) name the failure mode “instruction blindness”, while Zhan et al. (2026) describe a “modality collapse” towards a visual affordance prior. Mechanistically, Grant et al. (2026) find across six models and 394K episodes that injecting baseline activations into null-prompt episodes recovers near-identical behaviour.

Our position. We find the opposite on LIBERO-Goal: deleting the instruction destroys success at every scale. This does not contradict the work above, and the reconciliation is the point of this paper. The LIBERO suites differ in how much the scene reveals—`Spatial` varies positions of one object, `Object` varies objects within similar layouts, and `Goal` places *heterogeneous goals in a shared environment*. Only in the last does the observation fail to identify the task, so only there must a competent policy consult the instruction; consistent with this, blank-instruction results in the literature are strongly suite-dependent. Even there, redundancy has not disappeared—it has moved down a level, from the instruction as a whole to the verb within it. This matters because prior work increasingly proposes fixes: sharpness-aware finetuning (Zhang and Bisk, 2026), contrastive amplification of the language signal (Zhan et al., 2026), or suites built so language cannot be ignored (Hou and Zhao, 2026; Chen et al., 2026). Each presumes we can correctly measure how much a policy uses each part of an instruction, and we argue that measurement is not identifiable from perturbation experiments alone.

3 Experimental Setup

We conduct all evaluations on the LIBERO-Goal benchmarking suite in MuJoCo, chosen because its tasks share an object inventory and scene layout, so within a task the instruction is the only cue that disambiguates the goal. The invariant our causal claims require is per-task: for a given (task, episode), the initial state must be identi-

cal across all instruction conditions, $S_0(I_{\text{orig}}) \equiv S_0(I_{\text{pert}})$. We verify it by hashing the full simulator state (`qpos`, `qvel`) after reset and comparing across conditions; Section 4.3 reports what that check found. We evaluate three checkpoints chosen to vary both scale and action-head design: We evaluate **SmolVLA-0.45B** (Shukor et al., 2025), a compact SmolVLM2 backbone with a *flow-matching* action expert; **OpenVLA-7B** (Kim et al., 2024), a Prismatic VLM over Llama-2 7B emitting *discretized* action tokens; and **OpenVLA-OFT-7.5B** (Kim et al., 2025), the Optimized Fine-Tuning recipe (parallel decoding, 8-step action chunks, a continuous action representation and an L1-regression objective), which additionally consumes a wrist camera and 8-dimensional proprioception. (OFT here is *Optimized* Fine-Tuning, not the unrelated orthogonal-fine-tuning method; the public LIBERO-Goal release we evaluate is trained *without* FiLM, so we make no FiLM claims.) The three span a $16\times$ parameter range and all three major action parameterisations, which matters for Section 4.4: a finding that holds across discretized, continuous and flow-matching heads is unlikely to be an artifact of any one decoder.

Every condition is run under two seeds (7 and 42) with a rollout horizon of $T = 300$ steps, and each reported cell pools all episodes for its (model, condition) across both seeds.

Baseline fidelity. Because every number here is a difference from an `original` baseline, that baseline should be checked against the literature. Ours reproduces the published LIBERO-Goal result for OpenVLA-OFT closely (97.5% against 97.9% reported by Kim et al. (2025)) and runs below the published figures for the other two, at 78.5% for SmolVLA and 71.0% for OpenVLA-7B. We attribute the gap mainly to our fixed $T = 300$ step horizon: a visible share of episodes exhaust the budget rather than fail outright, whereas OpenVLA-OFT’s 8-step action chunks finish well inside it. We keep the shorter horizon deliberately, since it applies identically to every condition and every claim we make is a within-model contrast at a fixed horizon—a truncation removing the same tail from baseline and perturbed cells alike cannot create the differences we report, though it does mean our absolute rates should not be compared with leaderboard numbers.

Metrics

Let $\text{TSR}(c)$ be the Task Success Rate under condition c and TSR_0 the original baseline. The **Causal Sensitivity Score** $\text{CSS}(c) = (\text{TSR}_0 - \text{TSR}(c)) / \max(\text{TSR}_0, \epsilon)$ scores the destructive conditions: $\text{CSS} \approx 1$ means the instruction is causally necessary, ≈ 0 that it is ignored. The **Original-Action Rate (OAR)** is the fraction of semantic-swap episodes in which the policy still executes the *original* task despite the altered instruction; high OAR is direct evidence the policy follows pixels over words. For paraphrases we use **PRIDE** (Kim et al., 2026), which weights each paraphrase’s success by its distance from the original (we reuse the reference implementation; see Appendix C). Finally, **conditional task entropy** $H(\text{task} | W)$ is the residual uncertainty about which task an instruction denotes given only its words of class W —a property of the benchmark rather than of any model, used in Section 4.1 to establish which perturbations can carry evidence about grounding at all.

4 Empirical Results & Analysis

4.1 What the Instruction Actually Encodes

Perturbation studies infer which word classes a policy grounds by corrupting each in turn and reading off the damage. This is only valid if the word classes carry comparable information about the task. On **LIBERO-Goal** they do not, and the gap is total rather than marginal.

We parse each of the ten instructions into its unordered noun set and its verb set, and compute the residual uncertainty about task identity given each. With a uniform prior over the ten tasks, $H(\text{task}) = \log_2 10 = 3.32$ bits:

$$\begin{aligned} H(\text{task} | \text{noun set}) &= 0.00 \text{ bits}, \\ H(\text{task} | \text{verb set}) &= 1.55 \text{ bits}. \end{aligned} \quad (1)$$

All ten noun sets are distinct, so the nouns alone determine the task. The verbs do not: *put* is the verb of six of the ten tasks. Conditional on the nouns, the verb therefore contributes *exactly zero* bits. Figure 1 shows both the residual uncertainty and the partition that produces it.

Two consequences follow, and both undercut the standard reading of these probes. First, a policy that ignores the verb loses no task-relevant information on this suite: verb-insensitivity is optimal, not a grounding deficit. Second, and more

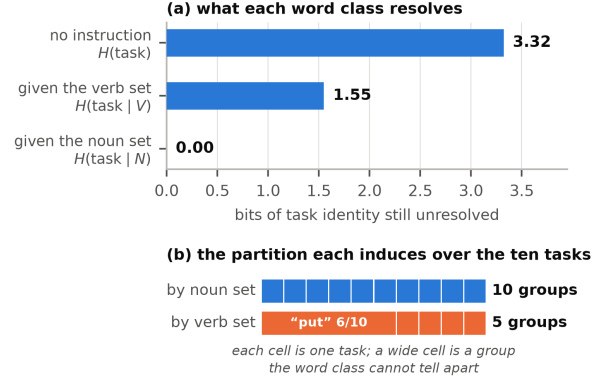


Figure 1: The instruction’s information structure, computed from the ten task definitions with no rollouts. (a) Task identity carries 3.32 bits; the noun set resolves all of them and the verb set leaves 1.55 unresolved. (b) Why: the noun sets partition the suite into ten singletons, while the verbs collapse six of the ten tasks into a single *put* group they cannot tell apart. A word class can only be probed for grounding to the extent that it separates tasks in the first place.

damaging for the method, the verb-substitution probe cannot detect verb-insensitivity even in principle—seven of its ten substitutions replace *put* with *push*, leaving the noun set and hence the task unchanged. Only three tasks (*open*↔*close* twice, *turn on*↔*turn off* once) substitute a true antonym where the nouns cannot disambiguate the goal; we report those separately throughout and treat them as the only valid verb probes the suite affords. This is a property of the benchmark, not of any model, computed from the task definitions with no rollouts, and we suggest it be reported as a matter of course before word-class grounding claims are drawn from perturbation experiments.

4.2 The Decisive Test: Removal, Not Substitution

The argument above makes a sharp prediction, and it is falsifiable. Instead of *substituting* within a word class—which, as just shown, mostly fails to change the task—*remove* a class outright and measure the cost. Dropping the verb (“the middle drawer of the cabinet”) removes 0 bits of task information and should cost close to nothing. Masking the nouns (“open the thing of the thing”) removes all 3.32 bits and should be catastrophic. Had verb removal instead collapsed performance, the redundancy account would be wrong: the information analysis would be describing the task distribution while missing what the policy actually

Condition	bits	TSR	Δ [95% CI]	p
<i>OpenVLA (7B)</i>				
original	0.00	71.0%	0.0	—
verb_dropped	0.00	75.0%	+4.0 [−3.5, +11.0]	0.36
nouns_masked	3.32	6.0%	−65.0 [−71.5, −58.0]	$<10^{-10}$
blank	3.32	0.0%	−71.0 [−77.0, −64.5]	$<10^{-10}$
<i>OpenVLA-OFT (7.5B)</i>				
original	0.00	97.5%	0.0	—
verb_dropped	0.00	95.5%	−2.0 [−5.0, +1.0]	0.34
nouns_masked	3.32	24.0%	−73.5 [−79.5, −67.5]	$<10^{-10}$
blank	3.32	12.5%	−85.0 [−89.5, −80.0]	$<10^{-10}$

Table 1: Word-class *removal*. **bits** is the task information the edit destroys, computed from the suite alone (Section 4.1); Δ is against a baseline restricted to the same episodes; every cell is $n = 200$. Coverage: OpenVLA (7B) on 10 of 10 tasks; OpenVLA-OFT (7.5B) on 10 of 10 tasks. The ablation conditions are generated in the same process as *original*, so the Δ column is a within-scene contrast.

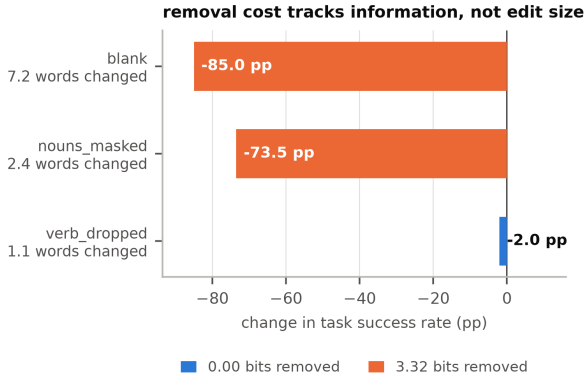


Figure 2: The removal test on OpenVLA-OFT ($n = 200$ per condition). Damage tracks the task information an edit destroys, not how much text it touches: deleting the verb changes 1.1 words and costs 2.0 points, while masking the nouns changes 2.4 and costs 73.5. “Words changed” is word-level edit distance from the original instruction, averaged over the ten tasks.

consumes.

The prediction holds, with an unusually clean separation (Table 1, Figure 2). Deleting the verb leaves OpenVLA-OFT’s success rate statistically unchanged on the same initial scenes: a point estimate of 2.0 points with a 95% interval of $[-5.0, +1.0]$ that contains zero, and a McNemar p of 0.34. Of 200 matched pairs only 10 changed outcome at all, 7 one way and 3 the other, which is what a coin looks like. Masking the nouns costs 73.5 points, flips 148 pairs, and never once flips a failure into a success. Because every perturbed episode starts from its baseline’s simulator state these are matched pairs, so the discordant counts are the evidence and the concordant majority correctly drops out. Two edits of comparable surface

magnitude differ by two orders of magnitude in effect; what separates them is not how much text was deleted but how much task information it carried.

One detail is worth noting: *nouns_masked* remains clearly *above* blank, even though on our accounting both destroy all 3.32 bits. The masked instruction retains the verb and the syntactic frame, so the residual gap estimates what a policy extracts from instruction *structure* once reference is gone—more than nothing, and much less than the nouns provide.

4.3 The Scene-Fixed Invariant Is Easy to Break Silently

Our protocol requires that all conditions for a (task, episode) start from an identical simulator state. Auditing this with the state hash, we found the invariant violated across most of our initial data, for a reason we believe generalises to other LIBERO-based evaluations: the initial state is *not* a pure function of (task, episode), because each *reset* advances the generator seeded once at construction, so the scene an episode receives depends on how many resets its process has already served. Any condition completed by a *second* process—after a wall-clock kill, or simply because conditions were run in separate jobs—silently lands on different scenes while the output files look ordered, complete and plausible (Appendix B).

This matters beyond bookkeeping: recomputing our paraphrase deltas against a scene-matched rather than a pooled baseline moves them by up to 12 percentage points, comparable to the effect sizes such studies report. We therefore (i) treat the task as the atomic unit of evaluation, so every condition for a task is produced by one process or regenerated wholesale, and (ii) make the invariant check a hard failure in aggregation rather than a logged diagnostic. Both are released with the code.

4.4 What the Causal Probes Show

Table 2 reports TSR, CSS and OAR under each structural and semantic corruption, with per-condition episode counts; Appendix A gives every cell with a 95% Wilson interval.

The destructive conditions behave as expected: blank and nonsense collapse success, establishing that the instruction is causally load-bearing. SmolVLA falls to exactly zero under both ($CSS = 1.00$); OpenVLA-OFT retains a small

Model	Baseline	Blank	Nonsense	Wrong Action	Wrong Object	Wrong Task	Repeated
SmolVLA (0.45B)	78.5%±2.5	0.0% (1.00)	0.0% (1.00)	64.5%±0.5	9.3%±2.1	0.0%	10.0%±2.0
OpenVLA (7B)	71.0%±1.0	0.0% (1.00)	0.0% (1.00)	59.0%	10.7%±0.7	0.0%	58.5%±0.5
OpenVLA-OFT (7.5B)	97.5%±1.5	12.5%±2.5 (0.87)	14.0%±1.0 (0.86)	94.5%±0.5	11.4%	0.0%	96.5%±1.5
<i>episodes per cell</i>	200	200	200	200	140	200	200

Table 2: TSR under each structural and semantic perturbation on LIBERO-Goal, with CSS in parentheses for the destructive conditions and $\pm$ the spread across seeds where the seeds disagree. Episode counts differ by condition and model, so there is no single N : original $n = 200$, blank $n = 200$, nonsense $n = 200$, wrong_action $n = 200$, wrong_object $n = 140$, wrong_task $n = 200$, repeated $n = 200$. Seeds 42, 7. All models pass the scene-fixed check on every comparable episode.

residue, 12.5% and 14.0% against a 97.5% baseline (CSS = 0.87 and 0.86), which we return to in the Discussion. Nothing here is a graded contrast, so the conclusion does not rest on the exact residue. The effect is flat in scale: CSS is 1.00, 1.00 and 0.87 across a $16\times$ parameter range, so sensitivity to losing the instruction tracks neither model size nor baseline competence. The two semantic-swap conditions are the ones that have been over-read, and we take them in turn.

What the scene-fixed audit lets us claim. A model whose grid fails the check (Section 4.3) is left blank rather than printed with a footnote. This is not hypothetical: every number reported here for OpenVLA-7B is from its *second* grid, the first having failed on 20 of 80 comparable episodes. All three grids now pass at 200 comparable episodes with zero mismatches. A scene confound can inflate or deflate a partial difference but cannot manufacture a floor, and wrong_task sits at exactly 0% for every model and episode, so the CSS ≈ 1 result is the robust part and the intermediate rates the fragile one. We lean only on the former.

Object swaps measure compliance, not collapse. Read naively, wrong_object triggers a near-total collapse in TSR. But TSR scores the policy against the *original* task, so a policy that correctly follows the altered instruction is recorded as a failure. The condition also mixes cases: the perturbed string may name another real task, describe a novel but coherent goal, or be degenerate (“put the bowl in the bowl”), where failure says nothing about grounding. The first is decisive. For one task the object swap yields, verbatim, another task’s instruction, and every model performs *that* task and never the original: SmolVLA 17/20, OpenVLA 18/20 and OpenVLA-OFT 20/20, with 0 executions of the original by any model. This is precise instruction-

Model	antonym		near-synonym	
	Δ TSR	p	Δ TSR	p
SmolVLA (0.45B)	−21.7 pp	10^{-3}	−10.7 pp	0.02
OpenVLA (7B)	−31.7 pp	10^{-6}	−3.6 pp	0.52
OpenVLA-OFT (7.5B)	−5.0 pp	0.45	−2.1 pp	0.51

Table 3: Drop in TSR under verb substitution, split by whether the substitution changes the task. Only the antonym column tests verb grounding; the near-synonym column is the control. p is an exact McNemar test over initial-state-matched pairs ($n = 60$ antonym, $n = 140$ near-synonym, pooled over seeds).

following, scored as total failure. We therefore report OAR against the *named* task alongside TSR, and caution that headline “collapse” numbers under object substitution should not be read as loss of grounding.

Verb swaps mostly do not test verbs. Seven of the ten verb substitutions preserve the noun set and therefore the task identity (Section 4.1), so surviving them is expected of any competent policy and says nothing about verb grounding. Splitting the condition into the three genuine antonym pairs and the seven near-synonyms separates the informative cases from the rest (Table 3).

The split changes the conclusion, and OpenVLA-7B shows why in the sharpest form (Figure 3). Read whole, its wrong_action condition costs 12 points—a mild effect, the kind routinely reported as evidence that verbs are near-optional. Split, that average dissolves into two opposite findings: the three antonym substitutions cost it 31.7 points ($p < 10^{-5}$, 19 of 60 pairs flipped to failure and none the other way), while the seven near-synonyms cost 3.6 and are indistinguishable from no edit at all ($p = 0.52$). OpenVLA reads the verb, decisively, and the suite-wide probe hides it by averaging one real effect against seven non-events.

SmolVLA behaves the same way, less starkly:

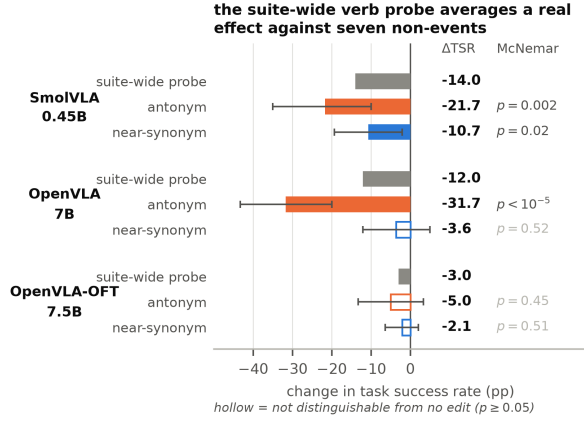


Figure 3: The suite-wide `wrong_action` probe against the two halves it averages. The pooled bar is the n -weighted mean of the other two ($n = 200$; 60 antonym episodes from the three tasks whose verb swap changes the goal, 140 near-synonym episodes from the seven where it does not), so for OpenVLA a -31.7 point effect and a null are reported as a mild -12.0 . Whiskers are 95% intervals over initial-state-matched pairs.

21.7 points on antonyms ($p = 0.002$) against 10.7 on near-synonyms ($p = 0.02$). OpenVLA-OFT is the exception, losing 5.0 points on antonyms and 2.1 on near-synonyms with *neither* distinguishable from zero ($p = 0.45$, $p = 0.51$); told to “turn off the stove” it turns the stove on, completing the *original* task in 55 of 60 episodes. The verb-insensitivity the literature attributes to VLAs in general is, on this suite, a property of one of the three models we tested—and the probe usually used to establish it could not have detected it in the other two.

The two 7B models also make a distinction visible that deserves care. *Removing* the verb is free for both—OpenVLA gains 4.0 points, OpenVLA-OFT loses 2.0, neither detectable (Table 1)—while *contradicting* it costs OpenVLA 31.7. Redundant is not the same as ignored: a word can carry no task information, so that deleting it changes nothing, and still be read closely enough that asserting its opposite derails the policy. Only comparing removal against substitution separates those two states.

This split remains the most underpowered comparison in the paper: the suite affords only three antonym pairs, so n grows with episodes but never with tasks. An earlier draft, built when each cell held two episodes, put OpenVLA-OFT’s antonym gap at exactly zero on $n = 12$; at $n = 60$ the estimate is 5.0 points, and the interval $[-13.3, +3.3]$

Model	Paraphrase Axis	n	scenes	TSR	Δ TSR (pp)	PRIDE
<i>SmolVLA (0.45B)</i>						
SmolVLA (0.45B)	Object	259	20	29.73%	-48.77	28.7
SmolVLA (0.45B)	Action	870	20	7.24%	-71.26	4.8
SmolVLA (0.45B)	Compositional	2963	20	1.79%	-76.71	1.5
SmolVLA (0.45B)	All axes	4092	24	4.72%	-73.78	2.7
<i>OpenVLA (7B)</i>						
OpenVLA (7B)	Object	150	150	48.00%	-23.00	36.3
OpenVLA (7B)	Action	150	150	60.67%	-10.33	51.8
OpenVLA (7B)	Compositional	150	150	24.00%	-47.00	20.2
OpenVLA (7B)	All axes	450	150	44.22%	-26.78	33.3
<i>OpenVLA-OFT (7.5B)</i>						
OpenVLA-OFT (7.5B)	Object	150	150	78.00%	-19.50	72.5
OpenVLA-OFT (7.5B)	Action	150	150	90.67%	-6.83	86.3
OpenVLA-OFT (7.5B)	Compositional	150	150	54.00%	-43.50	50.3
OpenVLA-OFT (7.5B)	All axes	450	150	74.22%	-23.28	65.8

Table 4: Paraphrase robustness. **scenes** is the number of distinct initial states the n episodes actually cover; where it is far below n , the episodes are repeated draws on the same scene and n overstates the evidence.

covers both. The lesson runs against the point estimates, not for them.

4.5 Paraphrase Robustness

Corrupting an instruction asks whether language is used at all. Paraphrasing it asks a harder question: whether the policy has bound to *meaning* or to surface form. We rewrite each instruction along three axes—the object expression, the action expression, and both at once—and score the results with PRIDE (Kim et al., 2026), which weights each paraphrase’s success by how far it departs from the original (Table 4).

The spread across models is large and orders with scale: SmolVLA degrades severely (4.72% overall TSR, PRIDE 2.7), OpenVLA sits in between (44.22%, PRIDE 33.3), and OpenVLA-OFT holds 74.22% (PRIDE 65.8). All three order the axes the same way—object paraphrases hurt least, compositional most—as expected if the noun phrase is what the policy keys on. We do not attribute OpenVLA-OFT’s robustness to FiLM: the checkpoint we evaluate is the plain OFT release (`use_film=false`), so that comparison is untested.

Two measurement caveats bound how these numbers should be read, both consequences of Section 4.3. First, Δ TSR compares a paraphrase pool against the *original* pool, and the two were produced by different processes, so scene overlap has to be checked rather than assumed; restricting the baseline to the matched subset shifts each delta by roughly ten percentage points. The *ordering* across axes (object < action < compositional) is identical under either baseline, so we treat the ordering as the result and the magnitudes as provisional, and ship both in the released CSV. Second, episode counts overstate the indepen-

dent variation behind these estimates: SmolVLA’s 2963 compositional paraphrase episodes span only two distinct initial scenes per task, because the paraphrase runs predate our initial-state cycling fix. We therefore report distinct scenes alongside n , and avoid confidence intervals computed as though episodes were independent draws.

4.6 When the Trajectories Part

Because perturbed and canonical rollouts share S_0 , their end-effector traces are comparable step by step. Defining t_{div} as the first step at which they differ by more than 5 cm, the kinematics match the causal results: with no usable instruction (`blank`, `nonsense`) the policy departs within the first step or two and never begins the right motion, whereas under `wrong_action` it begins the right motion and holds it for six to eleven steps before diverging—as expected when the noun set has already determined where to go and only the manner of acting is in question. We report this as a qualitative ordering rather than a precise timing, since pair counts per cell are small and t_{div} depends on the 5 cm threshold. Appendix D gives the curves and per-cell values.

5 Discussion and Conclusion

We presented a scene-fixed causal and paraphrase evaluation of three Vision-Language-Action models spanning $16\times$ in scale. Three things are worth separating in what we found.

Language is load-bearing, and scale does not change that. Destroying the instruction destroys success for all three architectures, across a $16\times$ range and across discretized, continuous and flow-matching action heads. Nothing suggests a bigger backbone learns to do without the instruction. This is the one headline that came through the re-analysis untouched.

The asymmetry we set out to explain was mostly ours to begin with. We began intending to give a mechanistic account of why VLAs bind nouns more tightly than verbs, and found that on this suite there is little to explain: the verb carries no task information once the nouns are known; seven of ten verb probes cannot detect verb-blindness even in principle; and most of the object-swap “collapse” is a policy doing what it was told, scored against a task it was not asked to do. The numbers are not in dispute—

our TSRs reproduce the pattern others report—only what may be concluded from them. What survives is narrower and better supported: on the only three tasks where a verb swap changes the goal, OpenVLA-7B loses 31.7 points ($p < 10^{-5}$) and SmolVLA 21.7 ($p = 0.002$), while OpenVLA-OFT shows no detectable response ($p = 0.45$). Verb-insensitivity is a property of one of these models, not of the class, and the suite-wide probe conventionally used to establish it reports a mild -12.0 points for the model it least describes.

Extra observation streams and residual short-cutting. OpenVLA-OFT is the only model here with a wrist camera and proprioception, and the only one retaining measurable success when the instruction is unusable (12.5% under `blank`, against exactly zero for the single-camera SmolVLA). Richer non-linguistic input enabling a visual shortcut is the most plausible reading, but with one model in that configuration we cannot separate the observation streams from the action head or the fine-tuning recipe, and do not claim to. It is not attributable to FiLM, which this checkpoint does not use.

What follows for the field. Instruction-perturbation studies inherit the information structure of their benchmark, and that structure has to be measured before behavioural results can be read as evidence about grounding. This bears directly on the mitigations now being proposed (Zhan et al., 2026; Zhang and Bisk, 2026). If a benchmark’s verbs are redundant given its nouns, a method that increases verb sensitivity is scored on a probe that cannot distinguish better grounding from a policy pushed away from the optimal one—so we do not dispute the reported gains, we question what they are measured against. What the field needs is suites built with verb-contrastive minimal pairs, of which LIBERO-Goal affords exactly three. Until they exist, the honest move is to publish the instruction set’s information content alongside the success rates.

Limitations

Our study is deliberately narrow to keep the causal claim clean, and this bounds its scope. (i) **Single suite:** the causal isolation relies on LIBERO-Goal’s shared initial scene and does not transfer to scene-confounded suites; we therefore do not claim the asymmetry holds under vi-

sual variation. (ii) **Simulation only**: all results are in MuJoCo/LIBERO with no real-robot evaluation. (iii) **Sample size**: several cells rest on a modest number of episodes, so we treat the qualitative pattern rather than any point estimate as the claim, and report n and a confidence interval for every cell so a reader can see which comparisons are underpowered. The antonym contrast is the narrowest of them: 60 episodes per model, but only ever three tasks, so episodes buy precision and never coverage—which is why we read it as a claim about one model rather than a trend. We note that going from two episodes per cell to ten moved OpenVLA-OFT’s antonym gap from 0.0 to 5.0 points, so we do not treat the remaining point estimates as settled either. (iv) **Scope of the information argument**: $H(\text{task} \mid W)$ is computed over this suite’s ten instructions under a uniform task prior. It shows that verbs are redundant *here*; it does not show that VLAs would ignore verbs on a suite where verbs are contrastive, and our three antonym pairs are too few to settle that. The uniform prior is itself an assumption: a policy trained on a skewed task distribution faces a different effective entropy than the one we compute from the task definitions. (v) **Coverage of the removal test**: the removal conditions were run on both 7B models and replicate ($n = 200$ each; verb deletion undetectable at $p = 0.36$ and $p = 0.34$, noun masking at $p < 10^{-10}$ for both). They were not run on SmolVLA, whose ~ 161 s per episode made two extra conditions cost roughly 18 GPU-hours we chose to spend on the causal grid instead; that omission is a budget decision, not a methodological one, and SmolVLA’s blank condition is consistent with the same account. (vi) **No mechanistic account**: we deliberately make no attention-level claim. An earlier draft reported a layer-wise noun-vs-verb cross-attention ratio; we found the underlying extraction had compared one model against a relabelled copy of another and had fed the OpenVLA-family models a blank image, so we removed the analysis rather than report it, and we leave a corrected measurement to future work rather than claim one here. Readers wanting a mechanistic picture should see Grant et al. (2026), who use activation injection, sparse autoencoders and linear probes across six models; our contribution is behavioural and information-theoretic, and is compatible with their finding that the visual pathway dominates on suites where the scene identifies the task. (vii) **Models**: we evalu-

ate three open checkpoints but a single benchmark family, and the coverage actually reported is whatever passes the scene-fixed check—see the draft-status note in Section 3. The $16\times$ scale range spanned by the models we report on is what carries the scale-invariance claim; it does not depend on the intermediate checkpoint.

Ethics and Reproducibility

This work uses only simulated manipulation tasks and public model checkpoints; it involves no human subjects or personal data. The primary societal consideration is safety: a policy that silently ignores the action verb could execute an unintended manipulation while appearing to follow instructions, which our diagnostics are designed to expose. For reproducibility we release per-run provenance (model, checkpoint hash, code revision, seed and wall-clock in a run manifest), pinned environment locks for each backbone, and the scene-fixed invariance audit. Every table in this paper is generated from the result files by a script rather than transcribed by hand; we adopted that rule after finding that hand-copied numbers were how an error survived several drafts of this work.

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A Full Per-Cell Results

Table 5 gives every RQ1 cell with its episode count and 95% Wilson interval, and Table 6 every kinematic-divergence cell. Both are generated from the result files rather than transcribed.

B How the Scene-Fixed Invariant Breaks

The LIBERO environment is seeded once at construction, and each `reset` advances that generator, which perturbs object poses even though the stored initial state pins `qpos` and `qvel`. A condition therefore inherits whatever generator position its process has reached. In our data the clearest case is a single condition split across a job boundary: episode 9 sits on a different scene from episodes 0–8 of the same condition. The remedy we adopt is to treat the task as the atomic unit of evaluation, so every condition for a task is produced by one process or regenerated wholesale, and to make the invariant check a hard failure in aggregation.

C PRIDE

PRIDE (Kim et al., 2026) weights each paraphrase’s success by how far it departs from the original instruction: $PD_i = 1 - (\alpha SK_i + (1 - \alpha)ST_i)$ and $PRIDE = 100 \sum_i s_i PD_i / \sum_i PD_i$, where s_i is success and SK, ST are keyword and structural similarities.

D Kinematic Divergence

Success rates say whether a rollout ended in the right place, not when it stopped following the original plan. Because our design gives the perturbed and canonical rollouts the same S_0 , we can compare their end-effector traces step by step and ask when they separate. We define t_{div} as the first step at which the two traces differ by more than 5 cm (Table 6, Appendix A).

Figure 4 shows the shape of this. The pattern matches the causal results. With no usable instruction (blank, nonsense) the policy departs from the canonical trajectory within the first step or two: it never begins the right motion. Under `wrong_action` it does begin the right motion

Model	Condition	n	TSR (95% CI)	CSS
SmolVLA (0.45B)	original	200	78.5% [72.3, 83.6]	–
SmolVLA (0.45B)	blank	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	nonsense	200	0.0% [0.0, 1.9]	1.00
SmolVLA (0.45B)	wrong_action	200	64.5% [57.7, 70.8]	–
SmolVLA (0.45B)	wrong_object	140	9.3% [5.5, 15.2]	–
SmolVLA (0.45B)	wrong_task	200	0.0% [0.0, 1.9]	–
SmolVLA (0.45B)	repeated	200	10.0% [6.6, 14.9]	–
OpenVLA (7B)	original	200	71.0% [64.4, 76.8]	–
OpenVLA (7B)	blank	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	nonsense	200	0.0% [0.0, 1.9]	1.00
OpenVLA (7B)	wrong_action	200	59.0% [52.1, 65.6]	–
OpenVLA (7B)	wrong_object	140	10.7% [6.6, 16.9]	–
OpenVLA (7B)	wrong_task	200	0.0% [0.0, 1.9]	–
OpenVLA (7B)	repeated	200	58.5% [51.6, 65.1]	–
OpenVLA-OFT (7.5B)	original	200	97.5% [94.3, 98.9]	–
OpenVLA-OFT (7.5B)	blank	200	12.5% [8.6, 17.8]	0.87
OpenVLA-OFT (7.5B)	nonsense	200	14.0% [9.9, 19.5]	0.86
OpenVLA-OFT (7.5B)	wrong_action	200	94.5% [90.4, 96.9]	–
OpenVLA-OFT (7.5B)	wrong_object	140	11.4% [7.2, 17.8]	–
OpenVLA-OFT (7.5B)	wrong_task	200	0.0% [0.0, 1.9]	–
OpenVLA-OFT (7.5B)	repeated	200	96.5% [93.0, 98.3]	–

Table 5: Every RQ1 cell with its episode count and 95% Wilson confidence interval, pooled over seeds 7 and 42. Read the intervals, not the point estimates: where n is small the `wrong_object` interval spans tens of percentage points, so we treat that condition as “near-total collapse” rather than as a precise rate shared across architectures. Every n in this table is generated from the result files rather than transcribed.

and holds it for six to eleven steps before diverging, depending on the model, which is what one expects when the noun set has already determined where to go and only the manner of acting is in question. We report this as a qualitative ordering rather than a precise timing: the pair counts per cell are small, and t_{div} depends on the 5 cm threshold.

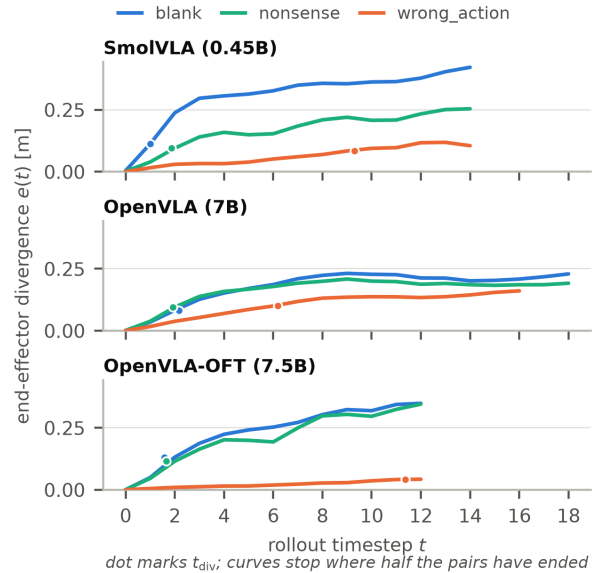


Figure 4: End-effector displacement $e(t)$ between the canonical rollout and each perturbed rollout from the same S_0 , for the three conditions discussed here. The destructive conditions leave the canonical path at once; `wrong_action` follows it and then peels away. Dots mark t_{div} . Each curve stops at the last step reached by at least half its pairs, so no part of it is an average over a handful of unusually long episodes.

Model	Condition	pairs	t_{div} (samples)	e_{10} [m]
SmolVLA (0.45B)	blank	84	1.00	0.3548
SmolVLA (0.45B)	nonsense	84	1.87	0.2197
SmolVLA (0.45B)	para_action	10	2.00	0.1972
SmolVLA (0.45B)	para_compositional	13	1.85	0.2086
SmolVLA (0.45B)	para_object	13	4.69	0.1383
SmolVLA (0.45B)	repeated	84	2.40	0.1737
SmolVLA (0.45B)	wrong_action	77	9.31	0.0838
SmolVLA (0.45B)	wrong_task	84	1.64	0.3278
SmolVLA (0.45B)	wrong_object	56	1.96	0.1785
OpenVLA (7B)	blank	73	2.16	0.2315
OpenVLA (7B)	nonsense	73	1.92	0.2087
OpenVLA (7B)	nouns_masked	73	3.11	0.2015
OpenVLA (7B)	repeated	65	6.35	0.1082
OpenVLA (7B)	verb_dropped	68	7.87	0.0764
OpenVLA (7B)	wrong_action	68	6.18	0.1355
OpenVLA (7B)	wrong_task	73	2.25	0.2992
OpenVLA (7B)	wrong_object	48	2.75	0.1920
OpenVLA-OFT (7.5B)	blank	55	1.58	0.3236
OpenVLA-OFT (7.5B)	nonsense	54	1.65	0.3043
OpenVLA-OFT (7.5B)	nouns_masked	55	2.67	0.2751
OpenVLA-OFT (7.5B)	para_action	50	11.70	0.0339
OpenVLA-OFT (7.5B)	para_compositional	55	6.58	0.1474
OpenVLA-OFT (7.5B)	para_object	51	10.90	0.0582
OpenVLA-OFT (7.5B)	repeated	43	12.02	0.0234
OpenVLA-OFT (7.5B)	verb_dropped	53	12.13	0.0206
OpenVLA-OFT (7.5B)	wrong_action	51	11.37	0.0289
OpenVLA-OFT (7.5B)	wrong_task	55	1.64	0.3127
OpenVLA-OFT (7.5B)	wrong_object	35	1.89	0.3173

Table 6: Kinematic divergence timestep t_{div} and end-effector displacement e_{10} at a fixed ten-step horizon. We quote e at ten steps rather than at the end of the rollout because pairs are truncated to the shorter of the two episodes and the median pair runs only 10–12 steps: every pair counted in **pairs** reaches step ten, so one n backs both columns, whereas an end-of-rollout figure would average the few longest—and so disproportionately failing—episodes.